

● EASY

● ALGEBRA

● ANSWER KEY

Linear functions

04

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Q1**B****B. 13**

Choice B is correct. It's given that the function f is defined by $fx = 3x - 8$. The value of $f7$ is the value of fx when $x = 7$. Substituting 7 for x in the given equation yields $f7 = 37 - 8$, which is equivalent to $f7 = 21 - 8$, or $f7 = 13$.

Choice A is incorrect. This is the value of $f7$ when $fx = 3x + 8$, rather than $fx = 3x - 8$.

Choice C is incorrect. This is the value of $f1$, rather than $f7$.

Choice D is incorrect. This is the value of $f-7$, rather than $f7$.

Q2**224**

The correct answer is 224. It's given that a student council group uses the function $px = 5x - 220$ to determine their profit px , in dollars, for selling x school posters. Substituting 900 for px in the given function yields $900 = 5x - 220$. Adding 220 to each side of this equation yields $1,120 = 5x$. Dividing each side of this equation by 5 yields $224 = x$. Therefore, in order to earn a profit of \$ 900, they must sell 224 school posters.

Q3**D**

D. $c = 25x + 11$

Choice D is correct. It's given that a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. If the rabbit's weight is x pounds, then multiplying 25 calories per pound by the rabbit's weight, x pounds, yields $25x$ calories. Adding the additional 11 calories that the rabbit should eat each day yields $25x + 11$ calories. It's given that c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds. Therefore, this situation can be represented by the equation $c = 25x + 11$.

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat $25 + 11$, or 36, calories per pound of the rabbit's weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 11 calories per pound of the rabbit's weight, plus an additional 25 calories.

Q4**D**

D. 30

Choice D is correct. It's given that when $x = 0$, $fx = 30$. Substituting 0 for x and 30 for fx in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of b is 30.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5

50

The correct answer is 50. It's given that the function f gives the total number of people on a company retreat with x managers. It's also given that 7 managers are on the company retreat. Substituting 7 for x in the given function yields $f7 = 77 + 1$, or $f7 = 50$. Therefore, there are a total of 50 people on a company retreat with 7 managers.

Q6

C

C. The object is moving at a rate of 16 inches per second.

Choice C is correct. It's given that in the equation $d = 16t$, d represents the distance, in inches, and t represents the number of seconds since an object started moving. In this equation, t is being multiplied by 16. This means that the object's distance increases by 16 inches each second. Therefore, the best interpretation of 16 in this context is that the object is moving at a rate of 16 inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of $16t$, rather than 16, in this context.

Choice D is incorrect and may result from conceptual errors.

Q7**A**

A. $h(x) = -x + 41$

Choice A is correct. An equation defining a linear function can be written in the form $hx = ax + b$, where a and b are constants. It's given that $h0 = 41$. Substituting 0 for x and 41 for hx in the equation $hx = ax + b$ yields $41 = a0 + b$, or $b = 41$. Substituting 41 for b in the equation $hx = ax + b$ yields $hx = ax + 41$. It's also given that $h1 = 40$. Substituting 1 for x and 40 for hx in the equation $hx = ax + 41$ yields $40 = a1 + 41$, or $40 = a + 41$. Subtracting 41 from the left- and right-hand sides of this equation yields $-1 = a$. Substituting -1 for a in the equation $hx = ax + 41$ yields $hx = -1x + 41$, or $hx = -x + 41$.

Choice B is incorrect. Substituting 0 for x and 41 for hx in this equation yields $41 = -0$, which isn't a true statement.

Choice C is incorrect. Substituting 0 for x and 41 for hx in this equation yields $41 = -410$, or $41 = 0$, which isn't a true statement.

Choice D is incorrect. Substituting 41 for hx in this equation yields $41 = -41$, which isn't a true statement.

Q8**C**

C. 60

Choice C is correct. It's given that t represents the number of seconds after the bus passes the marker. Substituting 2 for t in the given equation $d = 30t$ yields $d = 302$, or $d = 60$. Therefore, the bus will be 60 feet from the marker 2 seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker 1 second, not 2 seconds, after passing it.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the distance, in feet, the bus will be from the marker 3 seconds, not 2 seconds, after passing it.

Q9**C****C.** 175

Choice C is correct. The line shown represents the cost y , in dollars, for a manufacturer to make x rings. For the line shown, the x -axis represents the number of rings made by the manufacturer and the y -axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is represented by the y -coordinate of the point on the line that has an x -coordinate of 60. The point on the line with an x -coordinate of 60 has a y -coordinate of 175. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is 175.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make 20 rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make 100 rings.

Q10**A****A.** 10

Choice A is correct. It's given that the function f is defined by $fx = 5x + 8$. Substituting 58 for fx in this equation yields $58 = 5x + 8$. Subtracting 8 from both sides of this equation yields $50 = 5x$. Dividing both sides of this equation by 5 yields $10 = x$. Therefore, the value of x when $fx = 58$ is 10.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f58$, not the value of x when $fx = 58$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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